Alert Management Practice (Week 2)

1. Introduction
This document presents the implementation of SOC Alert Management Practice using a Security Information and Event Management (SIEM) system. The objective is to detect, classify, prioritize, and respond to security alerts using real system logs, MITRE ATT&CK mapping, and SOC risk scoring methodology.
The environment used is Ubuntu Linux monitored via Wazuh SIEM.

2. Tools Used
- Wazuh
- Ubuntu Linux (log source system)
- Google Sheets (classification + scoring)
- System logs: /var/log/auth.log

3. Observed Security Events
3.1 Authentication Failure Event
- Rule ID: 5503
- Rule Description: PAM: User login failed
- Event Type: Authentication Failure
- User: vboxuser
- MITRE ATT&CK: T1110.001 (Password Guessing)
- Timestamp: 2026-04-21 16:43:06 IST

Observation:
Single authentication failure detected. No repetition or escalation pattern observed.

3.2 Authentication Success Events
- Total Events Observed: 52
- Type: Successful authentication
- Status: Normal system behavior
- No anomalies detected

4. Alert Classification System

Alert ID	Event Type	Rule ID	MITRE Technique	Priority	Description
A001	PAM Login Failure	5503	T1110.001	Medium	Single authentication failure
A002	Authentication Success	-	-	Info	Normal system activity

5. SOC Risk Scoring Model (UPDATED INDUSTRY STANDARD)

5.1 Risk Factors

- Event severity (rule level)
- Event type
- Frequency
- Context (compromise vs normal behavior)

5.2 Risk Calculation

2 + 2 + 1 + 1 = 6

5.3 Risk Interpretation

Score	Severity
0–2	Low
3–5	Medium
6–7	Medium-High
8–10	Critical

Final Result:

- Authentication Failure → **6 (Medium-High Risk)**
- Authentication Success → **0–1 (Low / Informational)**

6. Incident Classification

Category	Classification
Authentication Failure	Credential Access (MITRE T1110.001)
Authentication Success	Normal System Activity

7. Incident Response Lifecycle (IMPROVED & COMPLETE)

7.1 Preparation Phase

- Wazuh SIEM configured for log monitoring
- Ubuntu system connected for event logging

7.2 Identification Phase

- PAM authentication failure detected via Wazuh
- Rule ID 5503 triggered
- Event mapped to MITRE T1110.001

7.3 Containment Phase

- No containment required (no compromise detected)
- System remained operational

7.4 Eradication Phase

- No malicious activity confirmed
- No remediation actions required

7.5 Recovery Phase

- System continued normal authentication operations
- 52 successful login events observed

7.6 Lessons Learned

- Single authentication failures are not sufficient indicators of attack
- Context and frequency are critical for SOC decision-making

8. Incident Ticket (SOC FORMAT)

Title:

[Medium-High Risk] PAM Authentication Failure Detected on Ubuntu Host

Details:

- User: vboxuser
- Log Source: /var/log/auth.log
- Rule ID: 5503
- MITRE: T1110.001
- SOC Risk Score: 6/10

Impact:

- No compromise detected
- No privilege escalation
- No brute-force pattern observed

Status:

Monitoring (No escalation required)

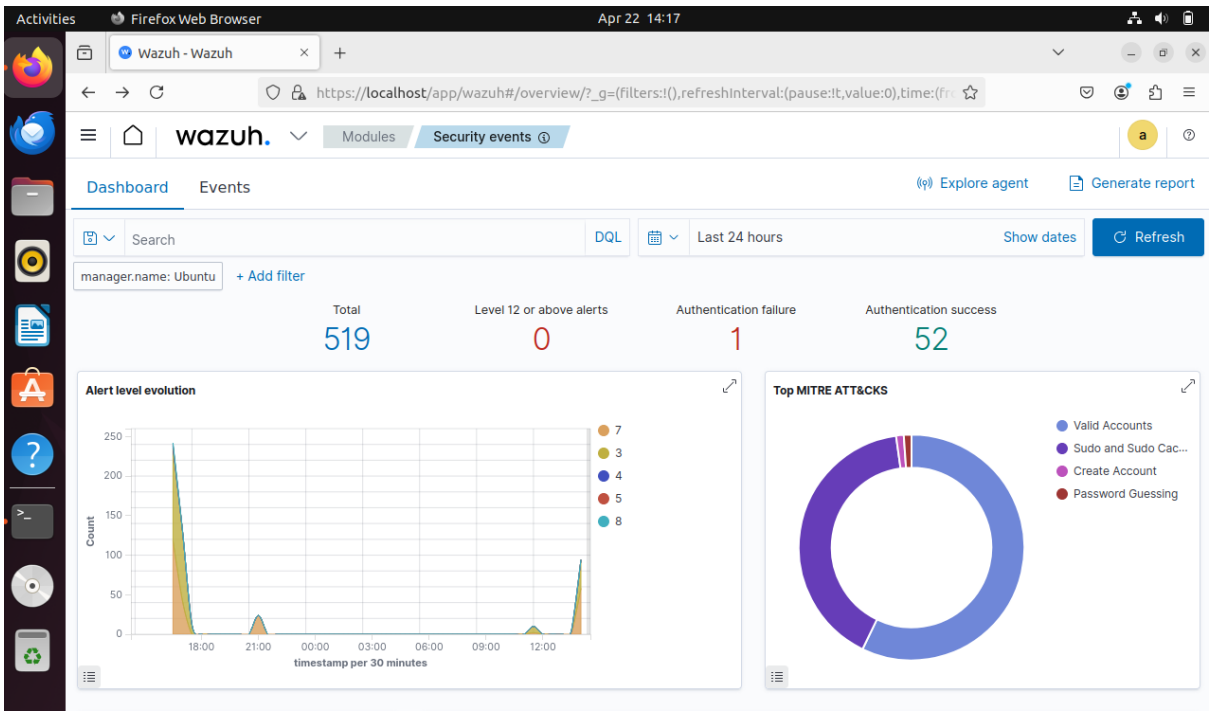
10. Conclusion

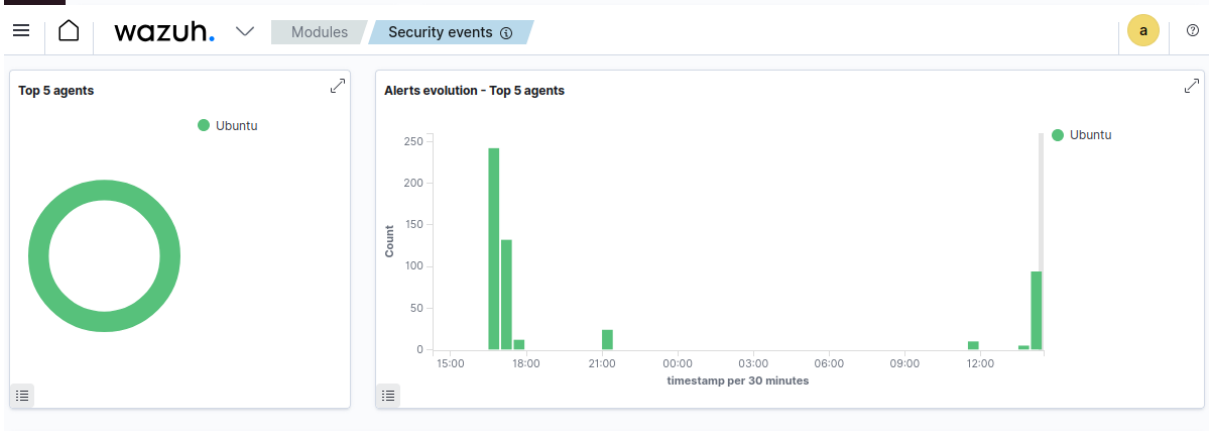
This practical demonstrates a complete SOC Alert Management workflow including:

- Real-time log analysis using Wazuh
- MITRE ATT&CK classification
- SOC risk scoring model (industry-standard approach)
- Incident lifecycle analysis
- Structured incident documentation
- SOC communication and escalation workflow

The analysis confirms that the detected event is a non-malicious authentication anomaly with no evidence of compromise.

Screenshots :





wazuh. Modules Security events

Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Apr 22, 2026 @ 14:06:16.477	000	Ubuntu	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	PAM: Login session opened.	3	5501
Apr 22, 2026 @ 14:03:50.278	000	Ubuntu			PAM: Login session closed.	3	5502
Apr 22, 2026 @ 14:03:44.245	000	Ubuntu			Dpkg (Debian Package) half configured.	7	2904
Apr 22, 2026 @ 14:03:44.245	000	Ubuntu			New dpkg (Debian Package) installed.	7	2902
Apr 22, 2026 @ 14:03:44.245	000	Ubuntu			Dpkg (Debian Package) half configured.	7	2904
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Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Apr 22, 2026 @ 14:03:44.244	000	Ubuntu			Dpkg (Debian Package) half configured.	7	2904
Apr 22, 2026 @ 14:03:44.244	000	Ubuntu			New dpkg (Debian Package) installed.	7	2902
Apr 22, 2026 @ 14:03:44.244	000	Ubuntu			New dpkg (Debian Package) installed.	7	2902
Apr 22, 2026 @ 14:03:44.243	000	Ubuntu			New dpkg (Debian Package) installed.	7	2902
Apr 22, 2026 @ 14:03:44.243	000	Ubuntu			Dpkg (Debian Package) half configured.	7	2904
Apr 22, 2026 @ 14:03:44.243	000	Ubuntu			Dpkg (Debian Package) half configured.	7	2904
Apr 22, 2026 @ 14:03:44.242	000	Ubuntu			Dpkg (Debian Package) half configured.	7	2904
Apr 22, 2026 @ 14:03:44.242	000	Ubuntu			New dpkg (Debian Package) installed.	7	2902
Apr 22, 2026 @ 14:03:44.242	000	Ubuntu			Dpkg (Debian Package) half configured.	7	2904
Apr 22, 2026 @ 14:03:44.242	000	Ubuntu			New dpkg (Debian Package) installed.	7	2902

```
vboxuser@Ubuntu: ~/soc-week2
vboxuser@Ubuntu:~$ sudo apt install apache2 -y
sudo systemctl start apache2
[sudo] password for vboxuser:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libsigsegv2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap
0 upgraded, 8 newly installed, 0 to remove and 4 not upgraded.
Need to get 1,933 kB of archives.
After this operation, 7,756 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64 1.7.0-8ubuntu0.22.04.2 [108 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 amd64 1.6.1-5ubuntu4.22.04.2 [92.8 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.2 [11.3 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.2 [9,170 B]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-bin amd64 2.4.52-1ubuntu4.19 [1,360 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-data all 2.4.52-1ubuntu4.19 [165 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-utils amd64 2.4.52-1ubuntu4.19 [89.5 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2 amd64 2.4.52-1ubuntu4.19 [97.9 kB]
Fetched 1,933 kB in 3s (636 kB/s)
Selecting previously unselected package libapr1:amd64.
(Reading database ... 332237 files and directories currently installed.)
Preparing to unpack .../0-libapr1_1.7.0-8ubuntu0.22.04.2_amd64.deb ...
Unpacking libapr1:amd64 (1.7.0-8ubuntu0.22.04.2) ...
Selecting previously unselected package libaprutil1:amd64.
Preparing to unpack .../1-libaprutil1_1.6.1-5ubuntu4.22.04.2_amd64.deb ...
Unpacking libaprutil1:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../2-libaprutil1-dbd-sqlite3_1.6.1-5ubuntu4.22.04.2_amd64.d
eb ...
```

```
vboxuser@Ubuntu:~$ for i in {1..20}; do curl http://localhost/?test=$i; done
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <!--
    Modified from the Debian original for Ubuntu
    Last updated: 2022-03-22
    See: https://launchpad.net/bugs/1966004
  -->
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Ubuntu Default Page: It works</title>
    <style type="text/css" media="screen">
  * {
    margin: 0px 0px 0px 0px;
    padding: 0px 0px 0px 0px;
  }

  body, html {
    padding: 3px 3px 3px 3px;

    background-color: #D8DBE2;

    font-family: Ubuntu, Verdana, sans-serif;
    font-size: 11pt;
    text-align: center;
  }

  div.main_page {
    position: relative;
    display: table;

    width: 800px;

    margin-bottom: 3px;
    margin-left: auto;
    margin-right: auto;
    padding: 0px 0px 0px 0px;

    border-width: 2px;
    border-color: #212738;
```

```
</html>
vboxuser@Ubuntu:~$ sudo apt install hydra -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libsigsegv2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  firebird3.0-common firebird3.0-common-doc libbson-1.0-0 libfbclient2
  libmemcached11 libmongoc-1.0-0 libmongocrypt0 libmysqlclient21 libpq5
  libserf-1-1 libsnappy1v5 libsvn1 libtommath1 libutf8proc2 mysql-common
Suggested packages:
  hydra-gtk
The following NEW packages will be installed:
  firebird3.0-common firebird3.0-common-doc hydra libbson-1.0-0 libfbclient2
  libmemcached11 libmongoc-1.0-0 libmongocrypt0 libmysqlclient21 libpq5
  libserf-1-1 libsnappy1v5 libsvn1 libtommath1 libutf8proc2 mysql-common
0 upgraded, 16 newly installed, 0 to remove and 4 not upgraded.
Need to get 4,452 kB of archives.
After this operation, 16.8 MB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 firebird3.0-common-doc all 3.0.8.33535.ds4-1ubuntu2 [26.8 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 firebird3.0-common all 3.0.8.33535.ds4-1ubuntu2 [15.5 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 libbson-1.0-0 amd64 1.21.0-1build1 [83.7 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtommath1 amd64 1.2.0-6ubuntu0.22.04.1 [56.5 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 libfbclient2 amd64 3.0.8.33535.ds4-1ubuntu2 [512 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libmemcached11 amd64 1.0.18-4.2ubuntu4 [88.2 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 libmongocrypt0 amd64 1.3.0-1ubuntu1 [93.9 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libsnappy1v5 amd64 1.1.8-1build3 [17.5 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 libmongoc-1.0-0 amd64 1.21.0-1build1 [311 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 mysql-common all 5.8+1.0.8 [7,212 B]
Get:11 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libmysqlclient21 amd64 8.0.45-0ubuntu0.22.04.1 [1,308 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpq5 amd64 14.22-0ubuntu0.22.04.1 [153 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 libserf-1-1 amd64 1.3.9-10ubuntu2 [50.0 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 libutf8proc2 amd64 2.7.0-3 [73.9 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libsvn1 amd64 1.14.1-3ubuntu0.22.04.1 [1,387 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 hydra amd64 9.2-1ubuntu1 [266 kB]
Fetched 4,452 kB in 2s (2,106 kB/s)
```

```
vboxuser@Ubuntu:~$ for i in {1..10}; do ssh fakeuser@localhost; done
ssh: connect to host localhost port 22: Connection refused
ssh: connect to host localhost port 22: Connection refused
ssh: connect to host localhost port 22: Connection refused
ssh: connect to host localhost port 22: Connection refused
ssh: connect to host localhost port 22: Connection refused
ssh: connect to host localhost port 22: Connection refused
ssh: connect to host localhost port 22: Connection refused
ssh: connect to host localhost port 22: Connection refused
ssh: connect to host localhost port 22: Connection refused
ssh: connect to host localhost port 22: Connection refused
vboxuser@Ubuntu:~$ mkdir soc-week2
cd soc-week2
nano incident.txt
vboxuser@Ubuntu:~/soc-week2$ nano escalation.txt
vboxuser@Ubuntu:~/soc-week2$ cat escalation.txt
Subject: Escalation - Brute Force Activity Detected

Multiple failed SSH login attempts detected on Ubuntu VM.
Wazuh has flagged authentication anomalies.
No successful login observed, but activity matches brute force pattern (T1110).

Recommendation:
Continue monitoring and block IP if repeated.

vboxuser@Ubuntu:~/soc-week2$
```